

Symmetric exclusion versus independent random walks via the Octopus inequality

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Abstract

We prove that the total-variation ε -mixing time of the k -particle symmetric exclusion process is at most 34 times the single-particle mixing time at error ε/k . The bound holds for every $\varepsilon \in (0, 1)$ on every finite connected graph with symmetric edge rates, and k may be replaced by the number of vacancies. This settles a conjecture of Oliveira. We also obtain bounds for the L^2 and L^∞ mixing times in terms of independent walks, and a Dirichlet-form inequality with the complete graph whose coefficient is optimal up to a universal constant. Further consequences include logarithmic Sobolev inequalities and energy bounds for the spin-1/2 Heisenberg ferromagnet.

1 Introduction

The symmetric exclusion process consists of particles performing random walks on a graph, with jumps to occupied vertices suppressed. Oliveira [26] conjectured that exclusion mixes at least as fast, up to a universal constant, as the corresponding independent walks. We establish a bound in terms of the single-particle mixing time that holds for every $\varepsilon \in (0, 1)$, uniformly over the graph, the symmetric edge rates and the number of particles. This bound implies Oliveira's conjecture and gives an estimate for the distance from equilibrium at every time.

Mixing times. Let V be a set of $n \geq 2$ vertices, and let $c_{xy} = c_{yx} \geq 0$ be edge rates, with $c_{xx} = 0$. Suppose that the graph formed by the edges of positive rate is connected. For $1 \leq k < n$, let $\text{EX}(k)$ denote the exclusion process on $\binom{V}{k}$, let $\text{RW}(1)$ denote the single-particle walk, and let $\text{RW}(k)$ denote the process of k independent labelled copies of this walk. These processes run in continuous time with the same edge rates, and their stationary distributions are uniform on $\binom{V}{k}$, V and V^k , respectively. Write $t^X(\varepsilon)$ for the worst-case ε -mixing time of a process X in total variation, and put $t_1(\varepsilon) = t^{\text{RW}(1)}(\varepsilon)$.

Our main result gives the following uniform bound.

Theorem 1.1. *Let V be a set of $n \geq 2$ vertices, and let $c_{xy} = c_{yx} \geq 0$ be edge rates. Suppose that the graph formed by the edges of positive rate is connected. Let $1 \leq k < n$ and $r = \min\{k, n - k\}$. Then*

$$t^{\text{EX}(k)}(\varepsilon) \leq 34 t^{\text{RW}(1)}(\varepsilon/r) \quad (0 < \varepsilon < 1). \quad (1.1)$$

For $0 < \varepsilon \leq 1/2$,

$$t^{\text{EX}(k)}(\varepsilon) \leq 25 t^{\text{RW}(1)}(\varepsilon/r). \quad (1.2)$$

Both processes use the edge rates c_{xy} .

We also obtain a bound for the distances from stationarity: writing $d_{\text{EX}(k)}(t)$ and $d_1(t)$ for the corresponding worst-case distances, we prove

$$d_{\text{EX}(k)}(34t) \leq \min\{1, rd_1(t)\} \quad (t \geq 0); \quad (1.3)$$

see Corollary 6.1.

Since $r \leq k$, Theorem 1.1 proves the bound asked for by Oliveira [26, Section 10, third question]:

$$t^{\text{EX}(k)}(\varepsilon) \leq Ct_1(\varepsilon/k) \quad (0 < \varepsilon < 1), \quad (1.4)$$

with a universal constant C . The argument ε/k reflects the product bound: if each of the k independent walks is within total variation distance ε/k of stationarity, their joint law is within distance ε . The reverse estimate in Proposition 2.4 then yields Oliveira's conjectured bound in terms of independent walks; see Corollary 6.2 for the explicit constants. The corresponding L^2 and L^∞ bounds are given in Corollary 6.3.

Earlier estimates established the order of the mixing time on several families of graphs. Lee and Yau [21] treated the cycle when the numbers of particles and vacancies both grow at least as a fixed positive power of n , Wilson [30] developed eigenfunction bounds for exclusion on a segment, and Morris [24] introduced the chameleon process to obtain the order of the mixing time on discrete tori of fixed dimension, including its dependence on the number of particles. Oliveira [26] extended the chameleon argument to arbitrary connected weighted graphs, giving an upper bound of order $t_1(1/4) \log(n/\varepsilon)$ for $0 < \varepsilon < 1/2$.

Hermon and Pymar [13] subsequently obtained bounds in terms of the single-particle spectral gap and spectral profile. For regular graphs of degree d with edge rates $1/d$, their general fixed-error bound is $O(\lambda^{-1} \log n \log \log n)$, where λ is the single-particle spectral gap. Taking $k \leq n/2$ by particle-hole symmetry, they proved Oliveira's bound for $n^\beta \leq k \leq n^{1-\beta}$, for every fixed $\beta \in (0, 1/2)$. For polynomially many particles, they also obtained it when $d \geq C \log_{n/k} n$ or $\lambda^{-1} \geq c(\log n)^4$, with the same normalisation of the rates. Their spectral-profile estimate further refines the dependence on k throughout $k \leq n^\delta$, for each fixed $\delta \in (0, 1)$.

For the interchange process with few labelled particles, Hermon and Pymar [14] obtained bounds in terms of independent walks on vertex-transitive graphs of fixed degree d with equal edge rates. For fixed $a \in (0, 1)$ and sufficiently large n , their result holds for $3 \leq k \leq n^a$ and errors at most $1/4$, with constants depending on a and d .

The relation with interchange gives another route to mixing-time bounds. The set of vertices occupied by any fixed collection of labels in the interchange process evolves as an exclusion process. Hermon and Salez [15] determined the order of the interchange mixing time on Cartesian products of connected graphs of a fixed size, including the hypercube: with unit edge rates, it is of order $\log |V|$. On these products the single-particle relaxation time is of constant order, and their Dirichlet-form inequality yields exclusion bounds for every number of particles.

Total-variation cutoff describes the sharp transition to equilibrium along growing families of graphs. Lacoïn and Leblond [20] proved cutoff on the complete graph, while Lacoïn proved cutoff on the segment [18] and determined the cutoff profile on the circle with half the vertices occupied [19]. Salez's lecture notes [28] discuss the relation between exclusion and independent walks in these questions. Hermon, Kozma and Salez [12] proved exclusion cutoff under assumptions on the particle density and the single-particle return probabilities. Under the same return-probability assumption, their Dirichlet-form inequality is asymptotically sharp at low density.

Spectral bounds. Hermon and Salez [15] place Oliveira's conjecture in the context of Aldous' spectral gap theorem. The motion of a single label in interchange is a random walk on the underlying graph,

and Aldous' theorem [6, Theorem 1.1 and Section 4.1.1] states that interchange, exclusion and this walk have the same spectral gap. At fixed total-variation error, the single-particle relaxation time bounds the mixing time up to a factor logarithmic in the number of vertices. Applying the common gap to exclusion started from a fixed configuration gives a bound with a factor $\log \binom{n}{k}$. Our second main result gives a Dirichlet-form inequality in terms of the single-particle spectral gap, with an optimal coefficient up to a universal constant.

Let L^{RW} and L_k^{EX} be the Laplacians of RW(1) and EX(k), respectively, and let λ be the spectral gap of L^{RW} . Let K_n denote the complete graph on V with unit rate on every edge, and write $\mathsf{L}_k^{\text{EX}}(K_n)$ for its exclusion Laplacian. Each Laplacian is the negative of the corresponding Markov generator. For self-adjoint operators, $S \succeq T$ means that $S - T$ is positive semidefinite; both exclusion Laplacians act on functions on $\binom{V}{k}$ with the uniform inner product.

Theorem 1.2. *Let V be a set of $n \geq 2$ vertices, and let $c_{xy} = c_{yx} \geq 0$ be edge rates. Suppose that the graph formed by the edges of positive rate is connected. Let λ be the spectral gap of the single-particle walk with these rates. For $1 \leq k < n$, let L_k^{EX} be the exclusion Laplacian with rates c_{xy} , and let $\mathsf{L}_k^{\text{EX}}(K_n)$ be the exclusion Laplacian on V with unit rates on every edge. Then*

$$\mathsf{L}_k^{\text{EX}} \succeq \frac{\lambda}{4n \log 6} \mathsf{L}_k^{\text{EX}}(K_n). \quad (1.5)$$

The coefficient has the best possible order: on the mean-zero single-particle subspace, $\mathsf{L}_1^{\text{EX}}(K_n) = n\mathbf{I}$, giving λ/n as an upper bound.

We also prove spectral lower bounds on the harmonic subspaces of the exclusion process. The space of functions on $\binom{V}{k}$ is an orthogonal sum of subspaces of degrees $0 \leq j \leq r$, each invariant under every transposition [10, Theorem 15]. Theorem 3.6 shows that the restriction of L_k^{EX} to the subspace of degree j has spectrum bounded below by $j\lambda/(4 \log 6)$. This subspace affords the irreducible representation indexed by $(n - j, j)$ [10, Section 4, following Lemma 18]. For $j \geq 1$, Aldous' theorem gives a lower bound of λ on its least eigenvalue; our bound grows linearly with the length j of the second row of the indexing partition. The relation between least eigenvalues in different irreducible representations is studied through the Aldous order introduced by Alon and Kozma [2].

Diaconis and Saloff-Coste [8] related exclusion to the Bernoulli–Laplace process by decomposing transpositions along paths, and Alon and Kozma [3] used the octopus inequality to relate the interchange Dirichlet form to that on the complete graph. Hermon and Salez [15, Section 2.1, Conjecture 2] conjectured that the single-particle relaxation time determines the optimal coefficient up to a universal constant. Theorem 1.2 proves the exclusion version of this conjecture for arbitrary symmetric edge rates.

For interchange, Hermon, Kozma and Salez [12, Proposition 3] proved a heat-kernel inequality on arbitrary finite weighted graphs. This is one of several consequences of the octopus inequality of Caputo, Liggett and Richthammer [6, Theorem 2.3], which arose from the network-reduction approach developed independently by Dieker [9] and by Caputo, Liggett and Richthammer. Earlier bounds for interchange generators were obtained from this inequality by Alon and Kozma [3] and Hermon and Salez [15]. Chen's moving particle lemma [7] is another consequence: it bounds the mean squared change under a transposition by the effective resistance between its endpoints times the exclusion Dirichlet form. The octopus inequality has also been related to universal Plücker coordinates and Schubert calculus [31].

Consequences. Corollary 6.4 gives an upper bound of order $\lambda^{-1} \log(n/\varepsilon)$ for the total-variation mixing time, uniformly in the number of particles. For regular expanders with total jump rate one at each vertex, the general fixed-error bound of Hermon and Pymar is $O(\log n \log \log n)$. Our estimate improves it by a

factor of $\log \log n$ and gives $O(\log n)$ for every number of particles, including the positive-density regime in Oliveira’s conjecture. Hermon and Pymar [13, Section 1.5] also discuss the distances from equilibrium of interchange and independent walks at the same time and from the same initial configuration, as well as an L^2 version of Oliveira’s conjecture, which we prove in Corollary 6.3.

Theorem 1.2 and the logarithmic Sobolev inequality of Lee and Yau [21, Theorem 5] give a lower bound of order $\lambda / \log(n^2/[k(n-k)])$ for the logarithmic Sobolev constant of exclusion; see Corollary 6.5. Thus this constant is comparable to λ when the numbers of particles and vacancies are both bounded below by a fixed positive proportion of n . Salez [27] obtained the corresponding conclusion for balanced colour classes under the conjectured Dirichlet-form inequality for interchange. Our theorem gives this conclusion for two colours on every connected weighted graph with symmetric edge rates.

The same spectral estimate applies to the spin-1/2 Heisenberg ferromagnet. Its standard correspondence with symmetric exclusion [25, Section 5.3, Proposition 2] identifies harmonic degree with the deficit of total spin from its maximum, giving an energy lower bound proportional to the product of this deficit and the single-particle spectral gap. The operator also occurs in quantum-cut sparsification, where Basu, Brakensiek, Kothari and Putterman [4] use an inequality with the complete graph to bound energies on expanders. Theorem 1.2 gives the corresponding operator inequality on every finite connected weighted graph with a coefficient optimal up to a universal constant. We give the statements and normalisations in Section 6.3.

Organisation. In Section 2, we define the processes and their mixing times, establish the single-particle and product estimates, and recall the heat-kernel inequality of Hermon, Kozma and Salez. In Section 3, we prove the spectral bound on each harmonic subspace and Theorem 1.2. In Section 4, we bound the harmonic components of the density and obtain a mixing-time estimate. In Section 5, we prove Theorem 1.1. In Section 6, we derive bounds for total-variation, L^2 and L^∞ mixing in terms of independent walks, together with logarithmic Sobolev inequalities and energy bounds for the Heisenberg ferromagnet.

Acknowledgements and use of generative AI. The analogy drawn by Hermon and Salez [15, Section 1.2] between Oliveira’s conjecture and Aldous’ spectral gap conjecture suggested to me the use of the octopus inequality. During this study, I used ChatGPT (versions 5.4, 5.5 and 5.6, accessed through ChatGPT Plus) to help me understand background material and identify standard terminology. ChatGPT also suggested an early argument for the heat-kernel inequality. The heat-kernel result used here follows from the more general interchange inequality of Hermon, Kozma and Salez. I am pretty grateful to them for this result. I also used ChatGPT to assist with checking references, proofreading and improving the English.

2 Preliminaries

We define the generators and distances used throughout the paper, then establish the product and single-particle bounds that relate the two formulations of Oliveira’s conjecture.

Throughout, let V be a set of $n \geq 2$ vertices. We assume that $c_{xy} = c_{yx} \geq 0$, $c_{xx} = 0$, and that the graph formed by the edges of positive rate is connected. Fix an ordering of V , so that sums over $x < y$ run over unordered pairs. All logarithms are natural.

The Laplacian of the single-particle walk is

$$(\mathbf{L}^{\text{RW}} f)(x) = \sum_{y \in V} c_{xy} (f(x) - f(y)). \quad (2.1)$$

The corresponding walk has transition kernel $P_t = e^{-tL^{\text{RW}}}$ and stationary distribution $\pi = \text{Unif}(V)$. Write λ for the smallest positive eigenvalue of L^{RW} , so that λ^{-1} is the relaxation time.

For $0 \leq r \leq n$, let $\Omega_r = \binom{V}{r}$ and $\pi_r = \text{Unif}(\Omega_r)$. We use τ_{xy} for both the transposition of x and y and its action on functions, $(\tau_{xy}f)(B) = f(\tau_{xy}B)$. The exclusion Laplacian is

$$L_r^{\text{EX}} = \sum_{x < y} c_{xy} (I - \tau_{xy}) \quad \text{on } L^2(\pi_r), \quad (2.2)$$

with transition kernel $Q_t^{(r)} = e^{-tL_r^{\text{EX}}}$. Thus L^{RW} and L_r^{EX} are the negatives of the corresponding Markov generators [6]. A transposition of two occupied or two empty vertices leaves the configuration unchanged, so each particle jumps from x to an unoccupied vertex y at rate c_{xy} , as in the single-particle walk.

For a probability measure σ on a finite set, write $\langle f, g \rangle_\sigma = \sum_x \sigma(x) f(x) g(x)$ and $\|f\|_{2,\sigma}^2 = \langle f, f \rangle_\sigma$. For probability measures μ, ν on the same set, write $\|\mu - \nu\| = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$ for their total variation distance; we use the same formula for signed measures of total mass zero. The worst-case distances from stationarity are

$$d_1(t) = \max_{x \in V} \|P_t(x, \cdot) - \pi\|, \quad (2.3)$$

$$d_{\text{EX}(r)}(t) = \max_{B \in \Omega_r} \|Q_t^{(r)}(B, \cdot) - \pi_r\|, \quad (2.4)$$

$$d_{\text{RW}(r)}(t) = \max_{\mathbf{x} \in V^r} \|P_t^{\otimes r}(\mathbf{x}, \cdot) - \pi^{\otimes r}\|. \quad (2.5)$$

The maximum in (2.5) includes tuples with repeated vertices. For a chain X , its total-variation mixing time $t^X(\varepsilon)$ is the infimum of the times at which its worst-case distance from stationarity is at most ε ; see [22]. We abbreviate $t_1(\varepsilon) = t^{\text{RW}(1)}(\varepsilon)$.

Since the complement map commutes with every transposition and sends π_r to π_{n-r} , particle-hole symmetry gives

$$d_{\text{EX}(r)}(t) = d_{\text{EX}(n-r)}(t) \quad (1 \leq r < n, t \geq 0). \quad (2.6)$$

Aldous' spectral-gap conjecture, posed in the early 1990s, asserted that interchange and the single-particle walk have the same gap. Handjani and Jungreis [11] proved the identity for weighted trees, while Koma and Nachtergaele [17] obtained related identities for spin chains. The conjecture was proved on every finite connected weighted graph by Caputo, Liggett and Richthammer [6]. The following theorem gives the corresponding spectral-gap identity for exclusion.

Theorem 2.1 ([6, Section 4.1.1, equation (4.1)]). *Let $1 \leq r < n$. Then the smallest positive eigenvalue of L_r^{EX} is λ . In particular, for every real function f on Ω_r ,*

$$\langle f, L_r^{\text{EX}} f \rangle_{\pi_r} \geq \lambda \text{Var}_{\pi_r}(f). \quad (2.7)$$

For a Markov semigroup K_t on a finite state space with strictly positive stationary distribution σ , its separation distance and total-variation contraction coefficient are

$$s_K(t) = 1 - \min_{x,y} \frac{K_t(x,y)}{\sigma(y)}, \quad \bar{d}_K(t) = \max_{x,y} \|K_t(x, \cdot) - K_t(y, \cdot)\|. \quad (2.8)$$

These are the continuous-time versions of the definitions in Levin and Peres [22]. Write $d_K(t)$ for its worst-case distance from σ . For the single-particle walk, we write $s_1(t)$ and $\bar{d}_1(t)$ for the corresponding separation distance and contraction coefficient. Since $K_t(x, \cdot) \geq (1 - s_K(t))\sigma$ for every x , we have $d_K(t) \leq s_K(t)$. The following lemma combines the standard contraction estimates [22, Lemma 4.10 and Exercise 4.3] with the factorisation of separation for product chains [13, Section 2.1, following equation (18)] and a reversible separation bound with constant 2.

Lemma 2.2. *Let K_t be a Markov semigroup on a finite state space with strictly positive stationary distribution. Then, for $t, u \geq 0$,*

$$\bar{d}_K(t) \leq 2d_K(t), \quad d_K(t+u) \leq d_K(t)\bar{d}_K(u). \quad (2.9)$$

If K_t is reversible, then

$$s_K(2t) \leq 2d_K(t). \quad (2.10)$$

For every integer $r \geq 1$, the product of r independent copies of the single-particle walk satisfies

$$1 - s_{\text{RW}(r)}(t) = (1 - s_1(t))^r. \quad (2.11)$$

Proof. To prove (2.10) for a reversible semigroup, put $f(z) = K_t(x, z)/\sigma(z)$ and $g(z) = K_t(y, z)/\sigma(z)$. Then

$$\begin{aligned} \frac{K_{2t}(x, y)}{\sigma(y)} &= \sum_z \sigma(z) f(z) g(z) \\ &\geq \sum_z \sigma(z) \min\{f(z), 1\} \min\{g(z), 1\} \\ &\geq 1 - \|K_t(x, \cdot) - \sigma\| - \|K_t(y, \cdot) - \sigma\|, \end{aligned}$$

where the last inequality follows from $ab \geq a + b - 1$ for $a, b \in [0, 1]$. Taking the minimum over x, y gives (2.10). $\square$

The next lemma gives two estimates at a prescribed single-particle mixing time: geometric decay at successive multiples of that time, and a lower bound in units of the relaxation time.

Lemma 2.3. *Let $0 < \delta < 1/2$ and $T = t_1(\delta)$. Then, for every integer $m \geq 1$,*

$$d_1(mT) \leq \delta(2\delta)^{m-1}, \quad \lambda T \geq \log \frac{1}{2\delta}. \quad (2.12)$$

Proof. Continuity gives $d_1(T) \leq \delta$, so iteration of (2.9) proves the first bound. For the second, apply the standard spectral lower bound at $t = T$ [22, Lemma 20.11, equation (20.34)]

$$d_1(t) \geq \frac{1}{2}e^{-\lambda t}. \quad (2.13)$$

$\square$

We now bound the mixing time of independent walks in terms of a single walk at the error used in Oliveira's formulation. The product estimate gives one inequality, and separation followed by two applications of contraction gives the other.

Proposition 2.4. Let $k \geq 1$ be an integer and $0 < \varepsilon \leq 1/4$. Then

$$t^{\text{RW}(k)}(\varepsilon) \leq t_1(\varepsilon/k) \leq 4t^{\text{RW}(k)}(\varepsilon). \quad (2.14)$$

Proof. The first inequality follows from the product bound [22, Exercise 4.4], $d_{\text{RW}(k)}(t) \leq kd_1(t)$. To prove the reverse bound, let $T = t^{\text{RW}(k)}(\varepsilon)$. The separation estimates in Lemma 2.2 give

$$d_1(2T) \leq 1 - (1 - 2\varepsilon)^{1/k} \leq \frac{-\log(1 - 2\varepsilon)}{k} \leq \frac{4(\log 2)\varepsilon}{k}.$$

Here the last inequality follows from convexity of $-\log(1 - 2x)$ on $[0, 1/4]$. On the other hand, projection onto one coordinate gives $d_1(T) \leq \varepsilon$ and hence $\bar{d}_1(T) \leq 2\varepsilon$. Two further applications of (2.9) now give

$$d_1(4T) \leq \frac{4(\log 2)\varepsilon}{k}(2\varepsilon)^2 \leq \frac{\varepsilon}{k}. \quad \square$$

At error $1/4$, the next proposition records the single-particle distance after eight independent-particle mixing times, together with a lower bound on the independent-particle mixing time in terms of the spectral gap.

Proposition 2.5. Let $1 \leq k \leq n$ and $T = t^{\text{RW}(k)}(1/4)$. Then

$$T \leq t_1(1/(4k)) \leq 4T, \quad (2.15)$$

$$d_1(8T) \leq \frac{1}{64k}, \quad T \geq \frac{\log(2k)}{4\lambda}. \quad (2.16)$$

Proof. Proposition 2.4 gives (2.15) and $d_1(4T) \leq 1/(4k)$. Since $\bar{d}_1(T) \leq 2d_1(T) \leq 1/2$, four applications of (2.9) then give $d_1(8T) \leq 1/(64k)$. Finally, comparing (2.13) at time $4T$ with $d_1(4T) \leq 1/(4k)$ yields the lower bound on T . $\square$

A heat-kernel inequality. We use the heat-kernel inequality of Hermon, Kozma and Salez, stated below for exclusion with symmetric edge rates. Its proof uses the octopus inequality of Caputo, Liggett and Richthammer [6, Theorem 2.3]. Earlier uses of the octopus inequality for interchange operators at successive time scales appear in Alon and Kozma [3] and Hermon and Salez [15].

For self-adjoint operators on the same real inner product space, write $S \preceq T$ if $\langle f, (T - S)f \rangle \geq 0$ for every f . We denote the all-ones vector by $\mathbf{1}$.

Definition 2.6. Let M be a real symmetric matrix indexed by V with $M\mathbf{1} = 0$, and let $0 \leq r \leq n$. Define the operator

$$\Phi_r(M) = - \sum_{x < y} M_{xy} (\mathbf{I} - \tau_{xy}) \quad \text{on } L^2(\pi_r). \quad (2.17)$$

The linear map Φ_r sends a graph Laplacian M to the exclusion Laplacian $\Phi_r(M)$ with edge rates $-M_{xy}$, so $\Phi_r(L^{\text{RW}}) = L_r^{\text{EX}}$. For the heat kernel $P_t = e^{-tL^{\text{RW}}}$, the operator $\Phi_r(\mathbf{I} - P_t)$ has edge rates $P_t(x, y)$ for $x \neq y$. The next theorem bounds this operator by a time multiple of the original exclusion Laplacian.

Theorem 2.7 ([12, Proposition 3, equation (40)]). *Let L^{RW} be the single-particle Laplacian in (2.1). Then, for every $t \geq 0$ and $0 \leq r \leq n$,*

$$\Phi_r(\mathbf{I} - e^{-tL^{\text{RW}}}) \preceq tL_r^{\text{EX}}. \quad (2.18)$$

The exclusion inequality follows by restricting the cited interchange inequality to functions of the image of an r -element set. To pass to arbitrary symmetric edge rates, choose $R \geq \max_x (L^{\text{RW}})_{xx} > 0$ and put $P = \mathbf{I} - L^{\text{RW}}/R$. This is a symmetric stochastic matrix, and the walk with generator $P - \mathbf{I}$ has transition kernel $e^{-tL^{\text{RW}}}$ at time Rt . Its exclusion Laplacian is L_r^{EX}/R , so the cited inequality at time Rt gives (2.18). Replacing counting measure by π_r multiplies both Dirichlet forms by the same constant. At $r = 0, n$, both operators vanish.

At a fixed multiple of the relaxation time, the spectral gap bounds the L^2 operator norm of the heat kernel on mean-zero functions. We apply the preceding inequality at this time and use the kernel bound to estimate the collision term arising from extension by zero in the next section.

3 Harmonic decomposition and spectral bounds

The harmonic decomposition expresses functions on Ω_k as sums of orthogonal components of different degrees and is determined by the permutation action on V , independently of the edge rates. We recall this decomposition and prove a spectral lower bound proportional to the degree. We apply Theorem 2.7 to the exclusion energy of a harmonic function and coordinatewise Poincaré inequalities to its extension by zero to ordered tuples. Write $(a)_j = a(a-1)\cdots(a-j+1)$, with $(a)_0 = 1$.

3.1 Harmonic decomposition

For $0 \leq r < n$, the raising and lowering operators between the layers Ω_r and Ω_{r+1} are

$$(\mathbf{R}_r f)(B) = \sum_{\substack{S \subset B \\ |S|=r}} f(S), \quad (\mathbf{D}_{r+1} g)(S) = \sum_{x \notin S} g(S \cup \{x\}).$$

Here f is a function on Ω_r and g is a function on Ω_{r+1} . We set $\mathbf{D}_0 = 0$ and interpret $\mathbf{R}_{-1}\mathbf{D}_0$ as zero. Throughout this subsection, adjoints and norms are taken with respect to counting measure unless stated otherwise; in particular, $\mathbf{R}_r^* = \mathbf{D}_{r+1}$.

Definition 3.1. For $0 \leq j \leq n/2$, the space of harmonic functions on Ω_j is $\mathcal{H}_j = \ker \mathbf{D}_j$. For $j \leq k \leq n$, the inclusion operator $\mathbf{U}_{j,k}$ is given by

$$(\mathbf{U}_{j,k} h)(B) = \sum_{\substack{S \subset B \\ |S|=j}} h(S), \quad (3.1)$$

and $\mathcal{H}_j^{(k)} = \mathbf{U}_{j,k} \mathcal{H}_j$. We call $\mathcal{H}_j^{(k)}$ the harmonic subspace of degree j on Ω_k .

In particular, $\mathcal{H}_0 = \mathbb{R}^{\Omega_0}$. For $j \geq 1$, a function h belongs to $\mathcal{H}_j$ precisely when

$$\sum_{x \notin S} h(S \cup \{x\}) = 0 \quad (S \in \Omega_{j-1}). \quad (3.2)$$

The inclusion operators intertwine the permutation actions on the respective layers, and repeated raising gives $\mathbf{U}_{j,k} = \mathbf{R}_{k-1} \cdots \mathbf{R}_j / (k-j)!$ for $k > j$. Under the identification of h with the multilinear polynomial $\sum_{S \in \Omega_j} h(S) \prod_{x \in S} z_x$, the condition $\mathbf{D}_j h = 0$ is the usual harmonicity condition for multilinear polynomials [10, Definition 1].

The following lemma collects the harmonic decomposition of Filmus and the raising and lowering relations and norm formula of Srinivasan. Each inclusion map is the iterated raising map divided by $(k-j)!$, and adjoints are taken in counting measure. Hermon, Kozma and Salez [12] use the same decomposition to study the spectrum of exclusion.

Lemma 3.2 ([10, Lemma 2 and Theorem 15]; [29, Theorem 1.2, equation (4)]). *Let $0 \leq r < n$. Then*

$$\mathbf{D}_{r+1} \mathbf{R}_r - \mathbf{R}_{r-1} \mathbf{D}_r = (n - 2r) \mathbf{I}. \quad (3.3)$$

For $0 \leq j \leq k \leq n/2$ and $h \in \mathcal{H}_j$,

$$\mathbf{U}_{j,k}^* \mathbf{U}_{j,k} h = \binom{n-2j}{k-j} h, \quad (3.4)$$

and consequently

$$\sum_{B \in \Omega_k} (\mathbf{U}_{j,k} h)(B)^2 = \binom{n-2j}{k-j} \sum_{S \in \Omega_j} h(S)^2. \quad (3.5)$$

Moreover, for $0 \leq k \leq n/2$,

$$L^2(\pi_k) = \bigoplus_{j=0}^k \mathcal{H}_j^{(k)}, \quad \dim \mathcal{H}_j^{(k)} = \binom{n}{j} - \binom{n}{j-1}, \quad (3.6)$$

where the sum is orthogonal and $\binom{n}{-1} = 0$.

The inclusion maps commute with the permutation action and hence intertwine the exclusion Laplacians on different layers. The following lemma gives the resulting equality of Rayleigh quotients.

Lemma 3.3 ([12, Theorem 2]). *Let $0 \leq j \leq k \leq n/2$. Then $\mathcal{H}_j$ and $\mathcal{H}_j^{(k)}$ are invariant under $\mathbb{L}_j^{\text{EX}}$ and $\mathbb{L}_k^{\text{EX}}$, respectively, and*

$$\mathbb{L}_k^{\text{EX}} \mathbb{U}_{j,k} = \mathbb{U}_{j,k} \mathbb{L}_j^{\text{EX}}. \quad (3.7)$$

For every non-zero $h \in \mathcal{H}_j$,

$$\frac{\langle \mathbb{U}_{j,k} h, \mathbb{L}_k^{\text{EX}} \mathbb{U}_{j,k} h \rangle_{\pi_k}}{\|\mathbb{U}_{j,k} h\|_{2, \pi_k}^2} = \frac{\langle h, \mathbb{L}_j^{\text{EX}} h \rangle_{\pi_j}}{\|h\|_{2, \pi_j}^2}. \quad (3.8)$$

The invariance and intertwining follow from the cited decomposition. The Rayleigh-quotient identity in our normalisation follows from (3.4): the inclusion map multiplies the energy and squared norm in counting measure by the same factor $\binom{n-2j}{k-j}$, and changing to uniform probability measures preserves both quotients.

Let $\mathbb{L}_k^{\text{EX}}(K_n) = \sum_{x < y} (\mathbb{I} - \tau_{xy})$ be the exclusion Laplacian on the complete graph with every edge rate equal to one. Each harmonic subspace is then an eigenspace. The eigenvalues in the following lemma, together with our spectral lower bound, give the complete-graph Dirichlet-form inequality.

Lemma 3.4 ([10, Lemma 18 and its proof; Lemma 24]). *Let $0 \leq j \leq k \leq n/2$. Then $\mathbb{L}_k^{\text{EX}}(K_n)$ acts on $\mathcal{H}_j^{(k)}$ as multiplication by $j(n-j+1)$.*

3.2 The spectral lower bound

For a symmetric stochastic matrix $\mathbf{P}$ on V , let $\mathbf{K} = \mathbb{I} - \mathbf{P}$ and let $\mathbf{K}^{(\ell)}$ act as $\mathbf{K}$ in coordinate ℓ of V^j , so that $\sum_{\ell=1}^j \mathbf{K}^{(\ell)}$ is the Laplacian of j independent walks with off-diagonal rates $\mathbf{P}(x, y)$. For $B \in \Omega_j$, write

$$r_{\mathbf{P}}(B) = \sum_{\substack{x, y \in B \\ x \neq y}} \mathbf{P}(x, y) \quad (3.9)$$

for the total rate of attempted jumps between occupied vertices; the sum is over ordered pairs.

A function h on Ω_j induces a symmetric function on the tuples in V^j with distinct coordinates. Its extension by zero to all of V^j is

$$F(x_1, \dots, x_j) = \begin{cases} h(\{x_1, \dots, x_j\}), & x_1, \dots, x_j \text{ are distinct,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.10)$$

The following identity relates the independent-particle energy of the zero extension to the exclusion energy and a collision term arising from attempted jumps between occupied vertices. We bound the collision term using the operator norm of the heat kernel.

Lemma 3.5. *Let $1 \leq j \leq n/2$, let $h \in \mathcal{H}_j$, and let F be its zero extension. Then F has mean zero in each coordinate with the others fixed, and*

$$\|F\|_{2, \pi^{\otimes j}}^2 = \frac{\binom{n}{j}}{n^j} \|h\|_{2, \pi_j}^2. \quad (3.11)$$

For every symmetric stochastic matrix $\mathbf{P}$ on V ,

$$\left\langle F, \sum_{\ell=1}^j \mathbf{K}^{(\ell)} F \right\rangle_{\pi^{\otimes j}} = \frac{\binom{n}{j}}{n^j} (\langle h, \Phi_j(\mathbb{I} - \mathbf{P})h \rangle_{\pi_j} + \mathbb{E}_{\pi_j}[r_{\mathbf{P}}(B)h(B)^2]). \quad (3.12)$$

Proof. Fix all but one coordinate. If a vertex is repeated among the fixed coordinates, F is identically zero as a function of the remaining coordinate. Otherwise, the fixed coordinates form a set $S \in \Omega_{j-1}$, and the harmonic condition gives $\sum_{x \notin S} h(S \cup \{x\}) = 0$ when we sum over the free coordinate. Thus the zero extension has mean zero in each coordinate with the other coordinates fixed. Since each j -element set has $j!$ orderings, (3.11) follows as well.

For the energy identity, consider a tuple with distinct coordinates and underlying set B . Jumps to unoccupied vertices contribute $(\Phi_j(\mathbf{I} - \mathbf{P})h)(B)$ to $\sum_{\ell=1}^j \mathbf{K}^{(\ell)} F$, whereas jumps to occupied vertices contribute $r_{\mathbf{P}}(B)h(B)$. Tuples with repeated coordinates contribute nothing to the inner product because F vanishes there. Grouping the $j!$ orderings of each set therefore gives (3.12), with the same factor $(n)_j/n^j$ as in (3.11). $\square$

We choose a time at which the single-particle heat kernel has a fixed spectral gap and apply coordinate-wise Poincaré inequalities to the zero extension. This bounds its independent-particle energy below, while the operator norm of the same kernel bounds the collision term above. Subtracting the collision term and applying Theorem 2.7 then bounds the exclusion energy.

Theorem 3.6. *Let $1 \leq j \leq k \leq n/2$ and $f \in \mathcal{H}_j^{(k)}$. Then*

$$\langle f, \mathbf{L}_k^{\text{EX}} f \rangle_{\pi_k} \geq \frac{j\lambda}{4 \log 6} \|f\|_{2, \pi_k}^2. \quad (3.13)$$

Proof. By Lemma 3.3, it suffices to prove the bound on $\mathcal{H}_j$. Let $u = (\log 6)/\lambda$ and $\mathbf{P} = \mathbf{P}_u$. The spectral gap of $\mathbf{I} - \mathbf{P}$ is $5/6$, so

$$\mathbf{P} \preceq \frac{1}{6} \mathbf{I} + \frac{5}{6n} \mathbf{1}\mathbf{1}^\top. \quad (3.14)$$

For $B \in \Omega_j$, applying this inequality to its indicator vector gives

$$r_{\mathbf{P}}(B) \leq \mathbf{1}_B^\top \mathbf{P} \mathbf{1}_B \leq \frac{j}{6} + \frac{5j^2}{6n} \leq \frac{7j}{12}, \quad (3.15)$$

where we used the non-negative diagonal entries of $\mathbf{P}$ and $j \leq n/2$.

Let $h \in \mathcal{H}_j$ and let F be its zero extension. Since F has mean zero in each coordinate with the other coordinates fixed, we may apply the Poincaré inequality for $\mathbf{I} - \mathbf{P}$ in each coordinate [22, Lemma 13.7] and sum to obtain

$$\left\langle F, \sum_{\ell=1}^j \mathbf{K}^{(\ell)} F \right\rangle_{\pi^{\otimes j}} \geq \frac{5j}{6} \|F\|_{2, \pi^{\otimes j}}^2.$$

Substituting (3.11) and (3.12), cancelling $(n)_j/n^j$, and subtracting the collision bound (3.15) gives

$$\langle h, \Phi_j(\mathbf{I} - \mathbf{P})h \rangle_{\pi_j} \geq \left(\frac{5j}{6} - \frac{7j}{12} \right) \|h\|_{2, \pi_j}^2 = \frac{j}{4} \|h\|_{2, \pi_j}^2. \quad (3.16)$$

Finally, Theorem 2.7 gives $\Phi_j(\mathbf{I} - \mathbf{P}) \preceq u \mathbf{L}_j^{\text{EX}}$, so dividing by u and applying (3.8) yields (3.13). $\square$

We prove the Dirichlet-form inequality stated in the introduction by combining the spectral bound on each harmonic subspace with its complete-graph eigenvalue. Here $\mathbf{L}_k^{\text{EX}}(K_n)$ is the exclusion Laplacian on the complete graph with unit edge rates.

Proof of Theorem 1.2. Suppose first that $k \leq n/2$. Both operators preserve the orthogonal decomposition in (3.6). On the subspace of degree j , Theorem 3.6 bounds $\mathbb{L}_k^{\text{EX}}$ below by $j\lambda/(4 \log 6)$, whereas Lemma 3.4 shows that $\mathbb{L}_k^{\text{EX}}(K_n)$ acts as $j(n-j+1)!$. The latter eigenvalue is at most nj for $j \geq 1$, and both operators vanish on constants; summing over the orthogonal subspaces therefore gives (1.5). The case $k > n/2$ follows by applying the complement map, which is an isometry intertwining the k -particle and $(n-k)$ -particle Laplacians for both graphs. $\square$

4 Inclusion probabilities and mixing

We combine the spectral decay from Theorem 3.6 with norm estimates obtained by conditioning on configurations containing no vertex whose occupation probability exceeds $2k/n$. Negative correlation for the unconditioned law bounds the inclusion probabilities of the conditional law after division by the probability of the conditioning event. An averaging operator then bounds the norms of the harmonic components, yielding Proposition 4.5.

4.1 Inclusion probabilities

The occupation probabilities evolve according to the single-particle heat equation, and negative correlation bounds inclusion probabilities by products of occupation probabilities. The following proposition records these two standard facts.

Proposition 4.1 ([23, Chapter VIII, Proposition 1.7]; [5, equation (17)]). Let $(B_t)_{t \geq 0}$ be the exclusion process started from a deterministic configuration $B_0 \in \Omega_k$, with occupation probabilities $m_t(x) = \mathbb{P}_{B_0}(x \in B_t)$. Then

$$m_t(x) = \sum_{a \in B_0} P_t(a, x). \quad (4.1)$$

For every $S \subseteq V$ and every $t \geq 0$,

$$\mathbb{P}_{B_0}(S \subseteq B_t) \leq \prod_{x \in S} m_t(x). \quad (4.2)$$

An empty product is interpreted as one.

We condition on the event that every occupied vertex has occupation probability at most $2k/n$. The next proposition bounds the probability of the discarded configurations by $2kd_1(t)$ and gives inclusion bounds for the conditional law.

Proposition 4.2. Let $1 \leq k \leq n/2$ and $t \geq 0$ satisfy $2kd_1(t) < 1$, and let μ be the law at time t of $\text{EX}(k)$ started from a deterministic configuration. Then there is an event $E \subseteq \Omega_k$, of probability $p = \mu(E) \geq 1 - 2kd_1(t) > 0$, such that $\nu = \mu(\cdot \mid E)$ satisfies

$$\nu\{B : S \subseteq B\} \leq p^{-1} \left(\frac{2k}{n} \right)^j \quad (S \in \Omega_j, 1 \leq j \leq k). \quad (4.3)$$

Proof. Let B_0 be the initial configuration and write $m(x) = \mu\{B : x \in B\}$. Using $\sum_x m(x) = k$ and (4.1), we have

$$\sum_x \left(m(x) - \frac{k}{n} \right)_+ = \frac{1}{2} \sum_x \left| m(x) - \frac{k}{n} \right| \leq \sum_{a \in B_0} \|P_t(a, \cdot) - \pi\| \leq kd_1(t), \quad (4.4)$$

where $a_+ = \max\{a, 0\}$. Let

$$D = \{x \in V : m(x) > 2k/n\}, \quad E = \{B \in \Omega_k : B \cap D = \emptyset\}.$$

Since $m(x) \leq 2(m(x) - k/n)$ for $x \in D$, the union bound and (4.4) give

$$1 - p \leq \sum_{x \in D} m(x) \leq 2kd_1(t). \quad (4.5)$$

It remains to bound the inclusion probabilities under ν . If S meets D , the probability is zero; otherwise, Proposition 4.1 gives

$$\nu\{B : S \subseteq B\} \leq p^{-1} \mu\{B : S \subseteq B\} \leq p^{-1} \prod_{x \in S} m(x) \leq p^{-1} (2k/n)^j. \quad \square$$

4.2 Bounds on harmonic components

The harmonic-component estimates will follow from an averaging operator whose quadratic form is expressed in terms of inclusion probabilities. Fix $1 \leq k \leq n/2$. For $0 \leq j \leq k$, let M_j be the transition kernel that first chooses uniformly a j -element subset of the current configuration and then chooses uniformly a k -element superset containing the chosen subset. Thus

$$M_j(B, C) = \frac{\binom{|B \cap C|}{j}}{\binom{k}{j} \binom{n-j}{k-j}}. \quad (4.6)$$

In particular, M_0 is the projection onto the constants and M_k is the identity. For each j , denote by Π_j the orthogonal projection in $L^2(\pi_k)$ onto $\mathcal{H}_j^{(k)}$.

The chain with transition kernel M_j is a special case of the Bernoulli–Laplace down-up chain studied by Khare and Zhou [16, Definition 4.16 and Theorem 4.19(2)], with each colour occurring once and $k - j$ balls moved at each of the two steps. The polynomial eigenspaces are the harmonic subspaces of the slice [10, Lemma 18], giving (4.7). The following lemma combines this eigenvalue with a quadratic-form identity for the inclusion probabilities.

Lemma 4.3. *Let $1 \leq k \leq n/2$ and $0 \leq j \leq k$. Then M_j is self-adjoint and positive semidefinite on $L^2(\pi_k)$. Its restriction to $\mathcal{H}_j^{(k)}$ is*

$$M_j|_{\mathcal{H}_j^{(k)}} = \frac{\binom{n-2j}{k-j}}{\binom{k}{j} \binom{n-j}{k-j}} \mathbf{I} = \frac{1}{\binom{k}{j}} \frac{(n-k)_j}{(n-j)_j} \mathbf{I}. \quad (4.7)$$

For any probability measure ν on Ω_k , with density $f = \nu/\pi_k$,

$$\langle f, M_j f \rangle_{\pi_k} = \frac{\binom{n}{j}}{\binom{k}{j}^2} \sum_{S \in \Omega_j} \nu\{B : S \subseteq B\}^2. \quad (4.8)$$

The self-adjointness and positivity can also be read from the factorisation, with the adjoint taken in counting measure,

$$M_j = \frac{U_{j,k} U_{j,k}^*}{\binom{k}{j} \binom{n-j}{k-j}}. \quad (4.9)$$

Proof of (4.8). Substitute $f(B) = \binom{n}{k} \nu(B)$ into the quadratic form of M_j and write

$$\binom{|B \cap C|}{j} = \sum_{S \in \Omega_j} 1_{\{S \subseteq B\}} 1_{\{S \subseteq C\}}.$$

Interchanging the sums gives (4.8), since $\binom{n}{k} / [\binom{k}{j} \binom{n-j}{k-j}] = \binom{n}{j} / \binom{k}{j}^2$. $\square$

We bound the quadratic form by the largest inclusion probability. Positivity and the scalar action on the subspace of degree j then give a bound on the squared norm of the corresponding harmonic component.

Proposition 4.4. Let $1 \leq k \leq n/2$, $p \in (0, 1]$, and let ν be a probability measure on Ω_k satisfying (4.3) for every $1 \leq j \leq k$. Then its density $f = \nu/\pi_k$ satisfies

$$\|\Pi_j f\|_{2, \pi_k}^2 \leq p^{-1} \frac{(4k)^j}{j!} \quad (1 \leq j \leq k). \quad (4.10)$$

Proof. Since M_j is self-adjoint and leaves $\mathcal{H}_j^{(k)}$ invariant, the quadratic form splits over this subspace and its orthogonal complement. By positive semidefiniteness and (4.7), we therefore have

$$\frac{1}{\binom{k}{j}} \frac{(n-k)_j}{(n-j)_j} \|\Pi_j f\|_{2, \pi_k}^2 \leq \langle f, M_j f \rangle_{\pi_k}. \quad (4.11)$$

To bound the right-hand side, observe that every configuration contains exactly $\binom{k}{j}$ subsets of size j , whence $\sum_{S \in \Omega_j} \nu\{B : S \subseteq B\} = \binom{k}{j}$. Thus, in each squared probability in (4.8), we bound one factor by (4.3) and sum the other to obtain

$$\langle f, M_j f \rangle_{\pi_k} \leq p^{-1} \frac{\binom{n}{j}}{\binom{k}{j}} \left(\frac{2k}{n} \right)^j.$$

Combining this with (4.11) gives

$$\|\Pi_j f\|_{2, \pi_k}^2 \leq p^{-1} \binom{n}{j} \left(\frac{2k}{n} \right)^j \frac{(n-j)_j}{(n-k)_j}. \quad (4.12)$$

It remains to estimate the falling-factorial ratio. Each denominator factor is positive and, for $0 \leq i < j$, we have $n-j-i \leq 2(n-k-i)$, since $n-2k+j-i \geq 0$. Hence the ratio in (4.12) is at most 2^j , and (4.10) follows using $\binom{n}{j} \leq n^j/j!$. $\square$

For a point-mass density, the squared norms of the harmonic components are the dimensions of the corresponding subspaces; see (6.5). By conditioning the law, Proposition 4.4 gives bounds in terms of the particle number. We let the conditional law evolve for a further interval, then combine the norm estimates with spectral decay and convexity to obtain the following mixing bound.

Proposition 4.5. Let $1 \leq k \leq n/2$ and $t, s \geq 0$. Then

$$d_{\text{EX}(k)}(t+s) \leq 2kd_1(t) + \frac{1}{2} \left[\exp \left(4ke^{-\lambda s/(2 \log 6)} \right) - 1 \right]^{1/2}. \quad (4.13)$$

Proof. The assertion is immediate if $2kd_1(t) \geq 1$, so suppose otherwise. Fix an initial configuration and let μ be the law at time t . By Proposition 4.2, there is a conditional law ν satisfying (4.3), with conditioning event of probability $p \geq 1 - 2kd_1(t) > 0$.

Let $f = \nu/\pi_k$. We have $\Pi_0 f = 1$, since ν is a probability measure, and by reversibility its density after a further time s is $e^{-sL_k^{\text{EX}}} f$. The harmonic subspaces are orthogonal and invariant, so Theorem 3.6 gives

$$\begin{aligned} \|e^{-sL_k^{\text{EX}}}(f - 1)\|_{2,\pi_k}^2 &= \sum_{j=1}^k \|e^{-sL_k^{\text{EX}}} \Pi_j f\|_{2,\pi_k}^2 \\ &\leq \sum_{j=1}^k \exp\left(-\frac{j\lambda s}{2\log 6}\right) \|\Pi_j f\|_{2,\pi_k}^2. \end{aligned} \quad (4.14)$$

By convexity, the L^2 bound on total variation distance [22, Section 4.7, equation (4.37)], and Proposition 4.4, we obtain

$$\begin{aligned} \|\mu Q_s^{(k)} - \pi_k\| &\leq 1 - p + \frac{p}{2} \|e^{-sL_k^{\text{EX}}}(f - 1)\|_{2,\pi_k} \\ &\leq 2kd_1(t) + \frac{\sqrt{p}}{2} \left[\sum_{j=1}^k \frac{(4k)^j}{j!} \exp\left(-\frac{j\lambda s}{2\log 6}\right) \right]^{1/2}. \end{aligned} \quad (4.15)$$

The result follows by using $p \leq 1$, extending the sum to all $j \geq 1$, and summing the exponential series. All bounds are uniform in the initial configuration. $\square$

5 Proof of the main theorem

We first apply Proposition 4.5 to bound the exclusion mixing time with an additive relaxation term. Using contraction and the single-particle spectral lower bound, we bound both terms by multiples of $t_1(\varepsilon/k)$. For $k = 2$ and ε close to one, we bound the mixing time directly by truncating the single-particle transition kernel.

5.1 General mixing estimates

Proposition 5.1. Let $1 \leq k \leq n/2$, $0 < \varepsilon < 1$ and $\tau \geq 0$ satisfy $d_1(\tau) \leq \varepsilon/(4k)$. Then

$$t^{\text{EX}(k)}(\varepsilon) \leq \tau + \frac{2\log 6}{\lambda} \log \frac{8k}{\varepsilon^2}. \quad (5.1)$$

In particular,

$$t^{\text{EX}(k)}(\varepsilon) \leq t_1\left(\frac{\varepsilon}{4k}\right) + \frac{2\log 6}{\lambda} \log \frac{8k}{\varepsilon^2}. \quad (5.2)$$

Proof. Apply Proposition 4.5 with $t = \tau$ and $s = (2\log 6)\lambda^{-1} \log(8k/\varepsilon^2)$. The assumption on the single-particle distance bounds the first term by $\varepsilon/2$. In the second term, the argument of the outer exponential is $\varepsilon^2/2$. Since $e^x - 1 \leq 2x$ for $0 \leq x \leq 1/2$, this term is also at most $\varepsilon/2$. Hence $d_{\text{EX}(k)}(\tau + s) \leq \varepsilon$, which proves the first assertion. Taking $\tau = t_1(\varepsilon/(4k))$ gives the second. $\square$

The next proposition estimates both additive terms by multiples of the single-particle mixing time, with a smaller constant for small errors and a bound for all errors when there are at least three particles.

Proposition 5.2. Let $2 \leq k \leq n/2$. Then for $0 < \varepsilon \leq 1/2$,

$$t^{\text{EX}(k)}(\varepsilon) \leq (3 + 12 \log 6) t_1(\varepsilon/k) < 25 t_1(\varepsilon/k). \quad (5.3)$$

If $3 \leq k \leq n/2$, then for every $0 < \varepsilon < 1$,

$$t^{\text{EX}(k)}(\varepsilon) \leq (5 + 16 \log 6) t_1(\varepsilon/k) < 34 t_1(\varepsilon/k). \quad (5.4)$$

Proof. Let $T = t_1(\varepsilon/k)$. Since $\varepsilon/k < 1/2$ in both ranges, Lemma 2.3 gives

$$d_1(mT) \leq \frac{\varepsilon}{k} \left(\frac{2\varepsilon}{k} \right)^{m-1}, \quad \lambda T \geq \log \frac{k}{2\varepsilon}. \quad (5.5)$$

If $k \geq 2$ and $\varepsilon \leq 1/2$, then $2\varepsilon/k \leq 1/2$ and hence $d_1(3T) \leq \varepsilon/(4k)$. Furthermore,

$$\begin{aligned} \log \frac{8k}{\varepsilon^2} &= 2 \log \frac{k}{2\varepsilon} + \log \frac{32}{k} \\ &\leq 2 \log \frac{k}{2\varepsilon} + \log 16 \leq 6 \log \frac{k}{2\varepsilon}, \end{aligned}$$

where the last step uses $k/(2\varepsilon) \geq 2$. Applying Proposition 5.1 with $\tau = 3T$ and using the second inequality in (5.5) gives (5.3).

Now suppose that $k \geq 3$ and $0 < \varepsilon < 1$. Using $2\varepsilon/k \leq 2/3$ and $(2/3)^4 < 1/4$, we obtain $d_1(5T) \leq \varepsilon/(4k)$. It remains to bound the relaxation term in the additive estimate. For this purpose, we use the inequality

$$\log \frac{8k}{\varepsilon^2} \leq 8 \log \frac{k}{2\varepsilon} \quad (5.6)$$

which is equivalent to $2048\varepsilon^6 \leq k^7$ and holds because $2048\varepsilon^6 < 2048 < 3^7 \leq k^7$. Applying Proposition 5.1 with $\tau = 5T$, followed by (5.6) and (5.5), gives (5.4). The constants in the statement follow from $3 + 12 \log 6 < 25$ and $5 + 16 \log 6 < 34$. $\square$

5.2 The two-particle case

For $k = 2$, we combine the lower bound $\lambda t_1(\varepsilon/2) \geq \log(1/\varepsilon)$ with truncation of the single-particle transition kernel to obtain a mixing bound uniform as $\varepsilon \uparrow 1$. The following truncation estimate is used to construct a subprobability measure dominated by the exclusion law, with mass bounded below and density with respect to the uniform stationary measure bounded above.

For a matrix R on V , write $Q(x, y) = (R(x, y) - 1/n)_+$.

Lemma 5.3. Let R be a doubly stochastic matrix on V such that

$$\max_{x \in V} \frac{1}{2} \sum_{y \in V} |R(x, y) - 1/n| \leq \frac{1}{2}.$$

Then for every integer $m \geq 1$,

$$R^m(x, y) \leq \frac{2}{n} + Q^m(x, y), \quad \sum_y Q^m(x, y) \leq 2^{-m}. \quad (5.7)$$

Proof. Since Q is non-negative and has row sums at most $1/2$, the row sums of Q^m are at most 2^{-m} . For the other inequality, note that $0 \leq (R - Q)(x, y) \leq 1/n$ entrywise. The telescoping identity

$$R^m - Q^m = \sum_{i=0}^{m-1} Q^i (R - Q) R^{m-1-i} \quad (5.8)$$

holds for arbitrary matrices R and Q . As R is doubly stochastic, we have $((R - Q)R^\ell)(x, y) \leq 1/n$ for every $\ell \geq 0$. Together with the bound on the row sums of Q^i , this shows that the i th summand in (5.8) is at most $2^{-i}/n$ entrywise. Summing the geometric series proves the first inequality in (5.7). $\square$

We apply the truncation estimate to the sixth power of the single-particle heat kernel. A bound on the row sums of the remainder gives a bound on the mass of configurations containing a vertex with large occupation probability. After discarding these configurations, we use negative correlation to bound the density of the remaining measure. This gives an overlap estimate and a bound after further evolution.

Lemma 5.4. *Let $n \geq 4$ and $T \geq 0$ satisfy $d_1(T) \leq 1/2$, and let μ be the law at time $6T$ of the two-particle exclusion process started from a deterministic configuration. Then there is a non-negative measure $g \leq \mu$ on Ω_2 such that*

$$g(\Omega_2) \geq \frac{27}{32}, \quad g \leq \frac{25}{2} \pi_2. \quad (5.9)$$

Consequently, uniformly in the initial configuration,

$$d_{\text{EX}(2)}(6T) \leq \frac{373}{400}, \quad (5.10)$$

$$d_{\text{EX}(2)}(6T + s) \leq \frac{5}{32} + \frac{5}{2\sqrt{2}} e^{-\lambda s} \quad (s \geq 0). \quad (5.11)$$

Proof. Let the initial particles be at a and b . Applying Lemma 5.3 to $R = P_T$ with $m = 6$, we find that the occupation probabilities at time $6T$ satisfy

$$m(y) = P_{6T}(a, y) + P_{6T}(b, y) \leq \frac{4}{n} + u(y), \quad \sum_y u(y) \leq \frac{1}{32},$$

where $u(y) = Q^6(a, y) + Q^6(b, y) \geq 0$. Let $D = \{y : m(y) > 5/n\}$ and $E = \{B \in \Omega_2 : B \cap D = \emptyset\}$. Since $y \in D$ implies $u(y) \geq m(y) - 4/n > m(y)/5$, the union bound gives

$$\mu(E^c) \leq \sum_{y \in D} m(y) \leq 5 \sum_y u(y) \leq \frac{5}{32}. \quad (5.12)$$

Let $g(C) = \mu(C)1_E(C)$. For $C = \{x, y\} \in E$, Proposition 4.1 gives $g(C) \leq m(x)m(y) \leq 25/n^2$, and since $\pi_2(C) = 2/[n(n-1)]$, we obtain $g(C) \leq (25/2)\pi_2(C)$. Together with (5.12), this proves (5.9).

Write $p = g(\Omega_2)$. Since $2g/25$ is bounded by both μ and π_2 , their overlap is at least $2p/25 \geq 27/400$; thus (5.10) follows from the identity $\|\mu - \pi_2\| = 1 - \sum_C \min\{\mu(C), \pi_2(C)\}$.

To prove (5.11), let $h = g/\pi_2$. Since $0 \leq h \leq 25/2$ and $\mathbb{E}_{\pi_2} h = p$, we have $\|h - p\|_{2, \pi_2}^2 \leq (25/2)p - p^2 \leq 25/2$. The remaining measure $\mu - g$ has mass $1 - p$, so its total-variation distance from $(1 - p)\pi_2$ is at most $1 - p$, also after evolution. Applying Theorem 2.1 to the centred density and using reversibility, we obtain

$$\begin{aligned} \|\mu Q_s^{(2)} - \pi_2\| &\leq 1 - p + \frac{1}{2} \|e^{-sL_{\text{EX}}^{(2)}}(h - p)\|_{2, \pi_2} \\ &\leq \frac{5}{32} + \frac{1}{2} e^{-\lambda s} \sqrt{\frac{25}{2}}, \end{aligned}$$

which is (5.11). $\square$

The overlap estimate gives a mixing-time bound for errors close to one; for smaller errors, the second bound in the preceding lemma applies after a further fixed multiple of the relaxation time. Combining the two bounds proves the following proposition.

Proposition 5.5. Let $n \geq 4$ and $1/2 < \varepsilon < 1$. Then

$$t^{\text{EX}(2)}(\varepsilon) < 34t_1(\varepsilon/2). \quad (5.13)$$

Proof. Let $T = t_1(\varepsilon/2)$. We have $d_1(T) \leq \varepsilon/2 < 1/2$, so Lemma 5.4 applies. For $\varepsilon \geq 15/16$, we have $373/400 < 15/16 \leq \varepsilon$ and hence $t^{\text{EX}(2)}(\varepsilon) \leq 6T$. This proves the required bound in this range. It remains to consider $1/2 < \varepsilon < 15/16$. Taking $s = (\log 6)/\lambda$ in (5.11) gives

$$d_{\text{EX}(2)}(6T + s) \leq \frac{5}{32} + \frac{5}{12\sqrt{2}} < \frac{1}{2} < \varepsilon.$$

Lemma 2.3 gives $\lambda T \geq \log(1/\varepsilon) > \log(16/15)$, and hence

$$t^{\text{EX}(2)}(\varepsilon) \leq 6T + \frac{\log 6}{\lambda} < \left(6 + \frac{\log 6}{\log(16/15)}\right) T < 34T.$$

The last inequality is equivalent to $6 < (16/15)^{28}$. This proves the bound in the remaining range. $\square$

Proof of Theorem 1.1. By particle-hole symmetry (2.6), it suffices to consider $k \leq n/2$, so that $r = k$. When $k = 1$, exclusion is the single-particle walk, so both conclusions hold. The first part of Proposition 5.2 gives (1.2) for $k \geq 2$ and $0 < \varepsilon \leq 1/2$. Its second part gives (1.1) for $k \geq 3$ and $0 < \varepsilon < 1$. It remains to consider $k = 2$ and $1/2 < \varepsilon < 1$, which follows from Proposition 5.5. $\square$

Since $\min\{k, n-k\} \leq k$ and t_1 is non-increasing in its error argument, Theorem 1.1 also gives $t^{\text{EX}(k)}(\varepsilon) \leq 34t_1(\varepsilon/k)$ for every $1 \leq k < n$. Thus it proves Oliveira's formulation (1.4) for all particle numbers.

6 Consequences

We bound the exclusion mixing times in total variation, L^2 and L^∞ in terms of those of independent particles, and also obtain a bound in terms of the relaxation time. We also derive a logarithmic Sobolev inequality and energy bounds for the Heisenberg ferromagnet from the complete-graph Dirichlet-form inequality.

6.1 Mixing-time bounds

The bound (1.1) can also be expressed in terms of the distance from stationarity at each time.

Corollary 6.1. Let $1 \leq k < n$ and $r = \min\{k, n-k\}$. Then

$$d_{\text{EX}(k)}(34t) \leq \min\{1, rd_1(t)\} \quad (t \geq 0). \quad (6.1)$$

Proof. The assertion is immediate if $rd_1(t) \geq 1$. Otherwise, let $rd_1(t) < \varepsilon < 1$. Since $t_1(\varepsilon/r) \leq t$, Theorem 1.1 gives $t^{\text{EX}(k)}(\varepsilon) \leq 34t$, and hence $d_{\text{EX}(k)}(34t) \leq \varepsilon$. Letting ε decrease to $rd_1(t)$ proves the bound. $\square$

Conversely, substituting $t = t_1(\varepsilon/r)$ into (6.1) gives (1.1), so the two formulations are equivalent.

Combining the main theorem with the product estimates from the preliminaries bounds the exclusion mixing time by a fixed multiple of the independent-particle mixing time at the same error. The two-time estimate gives a smaller constant at the usual fixed error.

Corollary 6.2. *Let $1 \leq k < n$ and $r = \min\{k, n - k\}$. Then*

$$t^{\text{EX}(k)}(\varepsilon) \leq 100 t^{\text{RW}(r)}(\varepsilon) \quad (0 < \varepsilon \leq 1/4). \quad (6.2)$$

At error $1/4$, we have the sharper bound

$$t^{\text{EX}(k)}(1/4) \leq (8 + 24 \log 6) t^{\text{RW}(r)}(1/4) < 52 t^{\text{RW}(r)}(1/4). \quad (6.3)$$

Proof. The first assertion follows by combining (1.2) with Proposition 2.4. To prove the second, we may assume $k \leq n/2$ by particle-hole symmetry. For $k = 1$ the two processes coincide, so suppose that $k \geq 2$ and let $T = t^{\text{RW}(k)}(1/4)$. By Proposition 2.5, we have $d_1(8T) \leq 1/(64k)$; applying Proposition 4.5 with $s = (2 \log 6)\lambda^{-1} \log(32k)$ therefore gives

$$d_{\text{EX}(k)}(8T + s) \leq \frac{1}{32} + \frac{1}{2} \sqrt{e^{1/8} - 1} < \frac{1}{4}.$$

Hence the mixing time is at most $8T + (2 \log 6)\lambda^{-1} \log(32k)$. Dividing by T and using the bound $T \geq \log(2k)/(4\lambda)$ from Proposition 2.5, together with $\log(32k) \leq 3 \log(2k)$ for $k \geq 2$, we obtain (6.3). $\square$

The same inclusion-probability and spectral estimates bound the L^2 mixing time of exclusion in terms of that of independent walks. The single-particle transition kernel gives a uniform bound on occupation probabilities, to which we apply the inclusion-probability estimate directly. By reversibility, the resulting bound also holds for L^∞ mixing.

For a process X , denote its transition kernel and stationary measure by Q_t^X and π^X , respectively. Define

$$d_p^X(t) = \max_z \left\| \frac{Q_t^X(z, \cdot)}{\pi^X} - 1 \right\|_{L^p(\pi^X)} \quad (p \in \{2, \infty\}),$$

and let $t_p^X(\varepsilon)$ be the least $t \geq 0$ for which $d_p^X(t) \leq \varepsilon$. Thus $d_2^X(t)^2$ is the worst-case χ^2 divergence from stationarity, and $d_\infty^X(t)$ is the worst-case pointwise relative error.

Corollary 6.3. *Let V be a set of $n \geq 2$ vertices, and let $c_{xy} = c_{yx} \geq 0$ be edge rates. Suppose that the graph formed by the edges of positive rate is connected. Let $1 \leq k < n$ and $r = \min\{k, n - k\}$. Then, for $p \in \{2, \infty\}$,*

$$t_p^{\text{EX}(k)}(\varepsilon) \leq 24 t_p^{\text{RW}(r)}(\varepsilon) \quad (0 < \varepsilon \leq 1).$$

Both processes use the edge rates c_{xy} .

Proof. By particle-hole symmetry, it suffices to consider $k \leq n/2$, and the case $k = 1$ is immediate. Suppose that $k \geq 2$, and let $T = t_2^{\text{RW}(k)}(\varepsilon)$ and $\delta = (1 + \varepsilon^2)^{1/k} - 1$. The product formula for the L^2 distance gives $d_2^{\text{RW}(1)}(T)^2 \leq \delta$. The spectral expansion also gives $e^{-2\lambda T} \leq \delta$, and $0 < \delta \leq \sqrt{2} - 1$.

Start exclusion from a fixed configuration, and let f be the density of its law at time $2T$ with respect to π_k . Reversibility and the Cauchy-Schwarz inequality give $n P_{2T}(x, y) \leq 1 + \delta < 2$. Proposition 4.1 therefore bounds the inclusion probability of each j -element set by $(2k/n)^j$. Applying Proposition 4.4 with the conditioning parameter p equal to one, we obtain

$$\|\Pi_j f\|_{2, \pi_k}^2 \leq \frac{(4k)^j}{j!} \quad (1 \leq j \leq k).$$

Let $s = 12(\log 6)T$. By (4.14),

$$\begin{aligned} \|e^{-s\mathbf{L}_k^{\text{EX}}} f - 1\|_{2,\pi_k}^2 &\leq \sum_{j=1}^k \frac{(4k)^j}{j!} e^{-j\lambda s/(2\log 6)} \\ &\leq e^{4k\delta^3} - 1 \leq (1 + \delta)^k - 1 = \varepsilon^2. \end{aligned}$$

For the last inequality, we used $4u^3 \leq \log(1 + u)$ for $0 \leq u \leq \sqrt{2} - 1$: indeed, $\log(1 + u) \geq u/(1 + u)$ and $4u^2(1 + u) \leq 12\sqrt{2} - 16 < 1$. The bounds are uniform in the initial configuration, and $2T + s < 24T$, proving the assertion for $p = 2$.

For a finite reversible chain, Levin and Peres [22, Proposition 4.15, equation (4.38)] give $d_\infty^X(2t) = d_2^X(t)^2$. Hence $t_\infty^X(\varepsilon) = 2t_2^X(\sqrt{\varepsilon})$, and the case $p = \infty$ follows with the same constant. $\square$

For the density of a point mass with respect to the stationary distribution, the squared norm of each harmonic component equals the dimension of the corresponding subspace. Applying the spectral estimate therefore gives a mixing bound which is uniform in the particle number.

Corollary 6.4. *Let $1 \leq k < n$ and $0 < \varepsilon < 1$. Then*

$$t^{\text{EX}(k)}(\varepsilon) \leq \frac{2\log 6}{\lambda} \log \frac{n}{\varepsilon^2}. \quad (6.4)$$

At error $1/4$, the right-hand side may be replaced by $(2\log 6)\lambda^{-1} \log(8n)$.

Proof. By particle-hole symmetry, we may assume that $k \leq n/2$. For $B \in \Omega_k$, let f_B be the density of the point mass at B with respect to π_k . We use the projection identity recorded by Hermon, Kozma and Salez [12, proof of Corollary 5, equations (70)–(72)], together with the dimension formula in Lemma 3.2:

$$\|\Pi_j f_B\|_{2,\pi_k}^2 = \dim \mathcal{H}_j^{(k)} = \binom{n}{j} - \binom{n}{j-1}. \quad (6.5)$$

Applying (4.14) to f_B , with the dimension bounded by $n^j/j!$, gives

$$d_{\text{EX}(k)}(t) \leq \frac{1}{2} \left[\exp \left(n e^{-\lambda t/(2\log 6)} \right) - 1 \right]^{1/2}. \quad (6.6)$$

For $t = (2\log 6)\lambda^{-1} \log(n/\varepsilon^2)$, the argument of the outer exponential is ε^2 . The convexity bound $e^x - 1 \leq (e - 1)x < 2x$ for $0 < x \leq 1$ now shows that the right-hand side of (6.6) is less than ε , as required. Taking $t = (2\log 6)\lambda^{-1} \log(8n)$ instead makes the same argument $1/8$, giving a distance strictly less than $1/4$. $\square$

Equations (6.5) and (4.10) show that conditioning replaces dependence on n by dependence on k in the harmonic-component bounds used for total variation.

6.2 Logarithmic Sobolev inequalities

Applying our complete-graph Dirichlet-form inequality in the argument of Hermon and Salez [15, Section 2.3, equation (30)] gives a logarithmic Sobolev inequality for exclusion with arbitrary symmetric rates. For a non-negative function g on Ω_k , define $\text{Ent}_{\pi_k}(g) = \mathbb{E}_{\pi_k}[g \log g] - \mathbb{E}_{\pi_k}[g] \log \mathbb{E}_{\pi_k}[g]$, with $0 \log 0 = 0$. We use the logarithmic Sobolev constant

$$\alpha_k = \inf_{\text{Ent}_{\pi_k}(f^2) > 0} \frac{\langle f, \mathbf{L}_k^{\text{EX}} f \rangle_{\pi_k}}{\text{Ent}_{\pi_k}(f^2)}.$$

Let $\alpha_k(K_n)$ be the corresponding constant for $\mathbb{L}_k^{\text{EX}}(K_n)$. Lee and Yau [21, Theorem 5] assign rate $1/n$ to each edge of the complete graph, and Salez uses the same rates with the reciprocal logarithmic Sobolev constant [27, Section 1.2]. Taking the reciprocal of the bound stated by Salez [27, Theorem 1.6] and rescaling the edge rates by n gives

$$\alpha_k(K_n) \geq \frac{n \log 2}{2 \log(n^2/[k(n-k)])}. \quad (6.7)$$

Theorem 1.2 gives $\alpha_k \geq \lambda \alpha_k(K_n)/(4n \log 6)$. The standard relation between the logarithmic Sobolev constant and the spectral gap [27, equation (1.6)], together with Theorem 2.1, gives $\alpha_k \leq \lambda/2$. Combining these bounds with (6.7), we obtain the following result.

Corollary 6.5. *Let V be a set of $n \geq 2$ vertices, and let $c_{xy} = c_{yx} \geq 0$ be edge rates. Suppose that the graph formed by the edges of positive rate is connected, and let $1 \leq k < n$. Then*

$$\frac{\lambda \log 2}{8 \log 6 \log(n^2/[k(n-k)])} \leq \alpha_k \leq \frac{\lambda}{2}. \quad (6.8)$$

In particular, for each $0 < \rho \leq 1/2$, the ratio α_k/λ is bounded above and below by positive constants depending only on ρ whenever $\rho n \leq k \leq (1 - \rho)n$.

When $k = n/2$, the lower bound becomes $\lambda/(16 \log 6)$. Thus Corollary 6.5 establishes the balanced two-colour conclusion discussed by Salez after Corollary 2.6 [27] for every connected weighted graph.

6.3 The Heisenberg ferromagnet

The classical correspondence between exclusion and the spin-1/2 Heisenberg ferromagnet [1, 25] allows us to express the spectral and operator bounds as energy inequalities. We use the following normalisation.

Let G denote the weighted graph on V with edge rates c_{xy} . Let $\mathbb{T}_{xy}$ be the operator on $(\mathbb{C}^2)^{\otimes V}$ that interchanges the tensor factors at x and y , and put $\mathbb{L}_G = \sum_{x < y} c_{xy}(\mathbb{I} - \mathbb{T}_{xy})$. On the invariant subspace with exactly k spins in the second basis state, the restriction of $\mathbb{L}_G$ is $\mathbb{L}_k^{\text{EX}}$. Hence $\mathbb{L}_G = \bigoplus_{k=0}^n \mathbb{L}_k^{\text{EX}}$, with trivial restrictions at $k = 0, n$, and applying Theorem 1.2 to each summand gives

$$\mathbb{L}_G \succeq \frac{\lambda}{4n \log 6} \mathbb{L}_{K_n}. \quad (6.9)$$

The restriction to the single-particle subspace shows that the coefficient cannot exceed λ/n , so the operator inequality has an optimal coefficient up to a universal constant.

In the normalisation used for quantum cuts by Basu, Brakensiek, Kothari and Putterman [4, Section 1.2], the two-site term $\mathbb{I} - \mathbb{T}_{xy}$ is twice the orthogonal projection onto the two-spin singlet subspace. Their proof of quantum-cut sparsification uses an inequality relating these operators to those on the complete graph [4, Theorem 5.5]. Inequality (6.9) holds on every finite connected weighted graph with a coefficient of order λ/n , uniformly over all quantum states, and provides the required complete-graph bound.

To express the result in the conventional Heisenberg normalisation, let $\mathbb{S}_x^a$ be one half of the a th Pauli matrix acting at x , for $a \in \{1, 2, 3\}$, and write $\mathbb{S}_x \cdot \mathbb{S}_y = \sum_{a=1}^3 \mathbb{S}_x^a \mathbb{S}_y^a$. The ferromagnetic Hamiltonian [25, Section 5.3, Proposition 2] with ground-state energy shifted to zero is

$$\mathbb{H}_G = \sum_{x < y} c_{xy} \left(\frac{1}{4} \mathbb{I} - \mathbb{S}_x \cdot \mathbb{S}_y \right) = \frac{1}{2} \mathbb{L}_G. \quad (6.10)$$

The squared total-spin operator is $S^2 = \sum_{a=1}^3 (\sum_x S_x^a)^2$. The Hamiltonian preserves the eigenspaces of this operator [25]. For $s \in \{n/2, n/2 - 1, \dots, n/2 - \lfloor n/2 \rfloor\}$, write $E_G(s)$ for the least eigenvalue of H_G on the eigenspace of S^2 with eigenvalue $s(s+1)$. The identity

$$\mathsf{L}_{K_n} = \frac{n(n+2)}{4} \mathsf{I} - S^2$$

together with Lemma 3.4 identifies total spin $n/2 - j$ with harmonic degree j in each subspace with a fixed number of particles. Applying Theorem 3.6 and particle-hole symmetry, and using the normalisation in (6.10), we obtain

$$E_G(n/2 - j) \geq \frac{j\lambda}{8 \log 6} \quad (0 \leq j \leq \lfloor n/2 \rfloor). \quad (6.11)$$

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